

- training a model by trial and error with giving the rewards for the expected output

Return

- multiply with some factor to the reward for every step
- the factor is called discount factor(γ)
- markov decision process (MDP) - for the formulation
- the first term is not discounted

State Action Value Function / Q Function

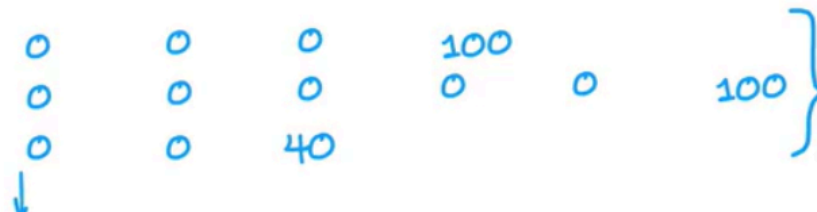
- $Q(s, a)$ = Return if you
 - start in state s
 - take action a (once)
 - then behave optimally after that

Bellman Equation

$$Q(s, a) = R(s) + \gamma \max_{a'} Q(s', a')$$

$$R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \dots$$

Stochastic Environment



The diagram shows a sequence of states and actions leading to multiple possible future states and rewards. A blue arrow points from the first state to the first action, which then branches into three possible next states. The first branch leads to a state with a reward of 100, the second to a state with a reward of 0, and the third to a state with a reward of 100. The third branch also includes a state with a reward of 40. A large blue bracket on the right groups the three possible next states and their rewards.

Expected Return = Average($R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \dots$)
 $= E[R_1 + \gamma R_2 + \gamma^2 R_3 + \gamma^3 R_4 + \dots]$

E-Greedy Policy

How to choose actions while still learning?

In some state s

Option 1:

Pick the action a that maximizes $Q(s, a)$.

Option 2:

- With probability 0.95, pick the action a that maximizes $Q(s, a)$. Greedy, "Exploitation"
- With probability 0.05, pick an action a randomly. "Exploration"

ϵ -greedy policy ($\epsilon = 0.05$)
0.95

$Q(s, \text{main})$ is low
↑
 a

Start ϵ high

Gradually decrease

Mini-Batch and Soft Update

Soft Update

Set $Q = Q_{\text{new}}$.

↑
 W, B

↑
 $W_{\text{new}}, B_{\text{new}}$

$$W = 0.01 W_{\text{new}} + 0.99 W$$

$$B = 0.01 B_{\text{new}} + 0.99 B$$